



COLLEGE  
OF  
MEDICINE

# Human immunodeficiency virus (HIV)

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# History of HIV

- ▶ Scientists believe that HIV originally came from a virus particular to chimpanzees <sup>(primates)</sup> in West Africa during the 1930s, and originally transmitted to humans through the transfer of blood through hunting.
- ▶ Over the decades, the virus spread through Africa, and to other parts of the world.
- ▶ However, it wasn't until the early 1980s, when rare types of pneumonia, cancer, and other illnesses were being reported to doctors that the world became aware of HIV and AIDS.
- ▶ This timeline highlights some of the major events and discoveries in HIV and AIDS since this time.

# Epidemiology

- The acquired immunodeficiency syndrome (AIDS) was first recognized in 1981
  - CDC publishes an article in its morbidity and mortality weekly report pneumocystis pneumonia- Los Angeles. The article describes cases of a rare lung infection, Pneumocystis carinii pneumonia (PCP), in five young, previously healthy gay men in Los Angeles.
- In 1983 it is named as LAV ( Lymphadenopathy associated virus)
- In 1986 International committee on taxonomy name a new name called Human Immunodeficiency Virus (HIV)

- AIDS is caused by HIV which progressively impairs cellular immunity
- HIV is a zoonotic infection with simian immunodeficiency viruses (SIV) from African primates, probably first infecting local hunters
- SIVs do not cause disease in their natural primate hosts
- HIV-1 was transmitted from chimpanzees and HIV-2 from sooty mangabey monkeys.



# Epidemiology

- HIV-1 is the cause of the global HIV pandemic,
- while HIV-2, which causes a similar illness to HIV-1 but progresses more slowly and is less transmissible, is restricted mainly to western Africa

# Epidemiology

- There are three groups of HIV-1, representing three separate transmission events from chimpanzees:
  - M ('major', worldwide distribution),
  - O ('outlier')
  - N ('non-major and non-outlier').
- Groups O and N are restricted to West Africa.
- Group M consists of nine subtypes: A-D, F-H, J and K, but recombinants of subtypes occur frequently

# Epidemiology

- Globally, subtype C (which predominates in sub-Saharan Africa and India) accounts for half of infections and appears to be more readily transmitted.
- Subtype B predominates in Western Europe, the Americas and Australia.
- In Europe, the prevalence of non-B subtypes is increasing because of migration.
- Subtypes A and D are associated with slower and faster disease progression, respectively.

# Global situation ( 2024 data)

- ▶ 91.4 million [73.4 million–116.4 million] people have become infected with HIV since the start of the epidemic.
- ▶ 44.1 million [37.6 million–53.4 million] people have died from AIDS-related illnesses since the start of the epidemic.
- 40.8 million [37.0 million–45.6 million] people globally were living with HIV in 2024.

# Global prevalence ( 2024)

- ▶ 39.4 million [35.7 million–44.0 million] adults (15 years or older).
- ▶ 1.4 million [1.1 million–1.8 million] children (0–14 years). */ Vertical transmission*
- ▶ 53% of all people living with HIV were women and girls.
- ▶ 87% [69–>98%] of all people living with HIV knew their HIV status in 2024.
- ▶ About 5.3 million people did not know that they were living with HIV in 2024. *silently spread it long incubation period but relatively short latent period*

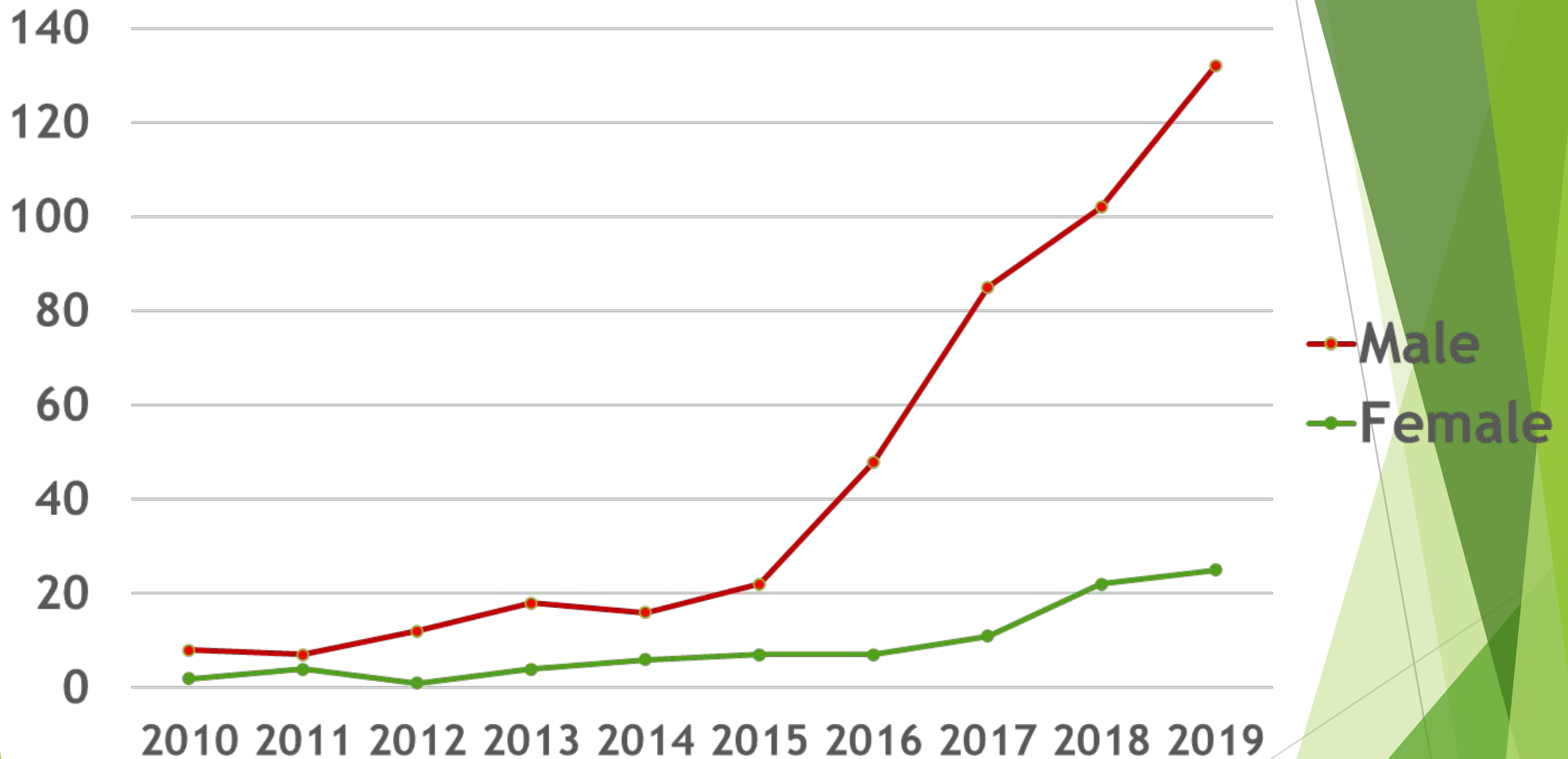
# New cases 2024 ( incidence)

- ▶ In 2024, 1.3 million [1 million–1.7 million] people were newly infected with HIV, compared to 3.4 million [2.7 million–4.3 million] people in 1996.
- ▶ Women and girls accounted for 45% of all new infections in 2024.
- ▶ Since 2010, new HIV infections have declined by 40%, from 2.2 million to 1.3 million in 2024. However, this falls short of the target of getting below 370 000 new infections by 2025.

HIV & AIDS is one of those diseases that we consider both incidence & prevalence → how many total case we have → extent of disease  
→ for chronic  
→ to see how many new case we have :- endemic → epidemic/ pandemic  
→ for acute disease

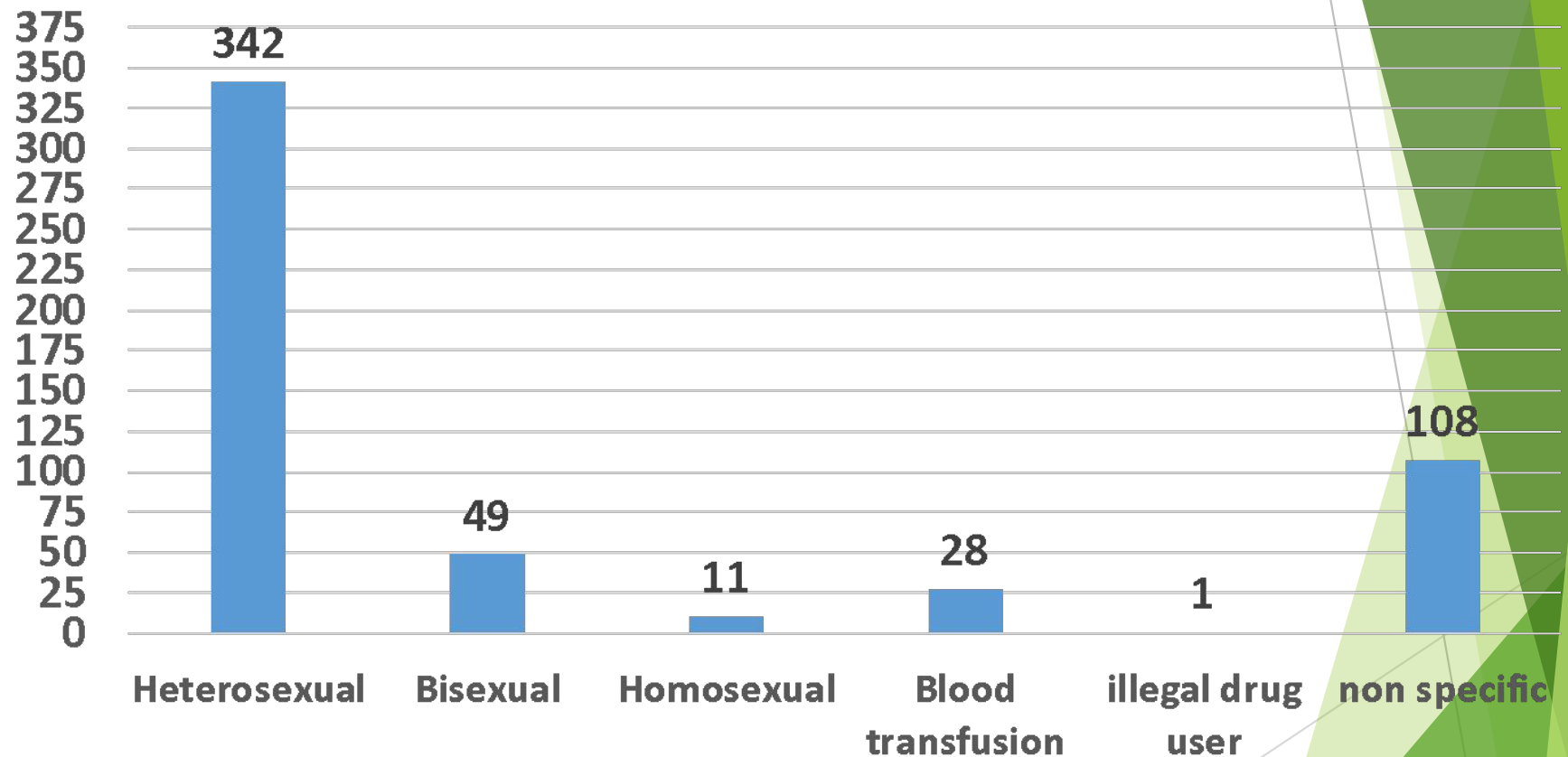


# Recorded HIV cases in Iraq





# Possible routes of entry of HIV Iraqi cases (2010 - 2019)

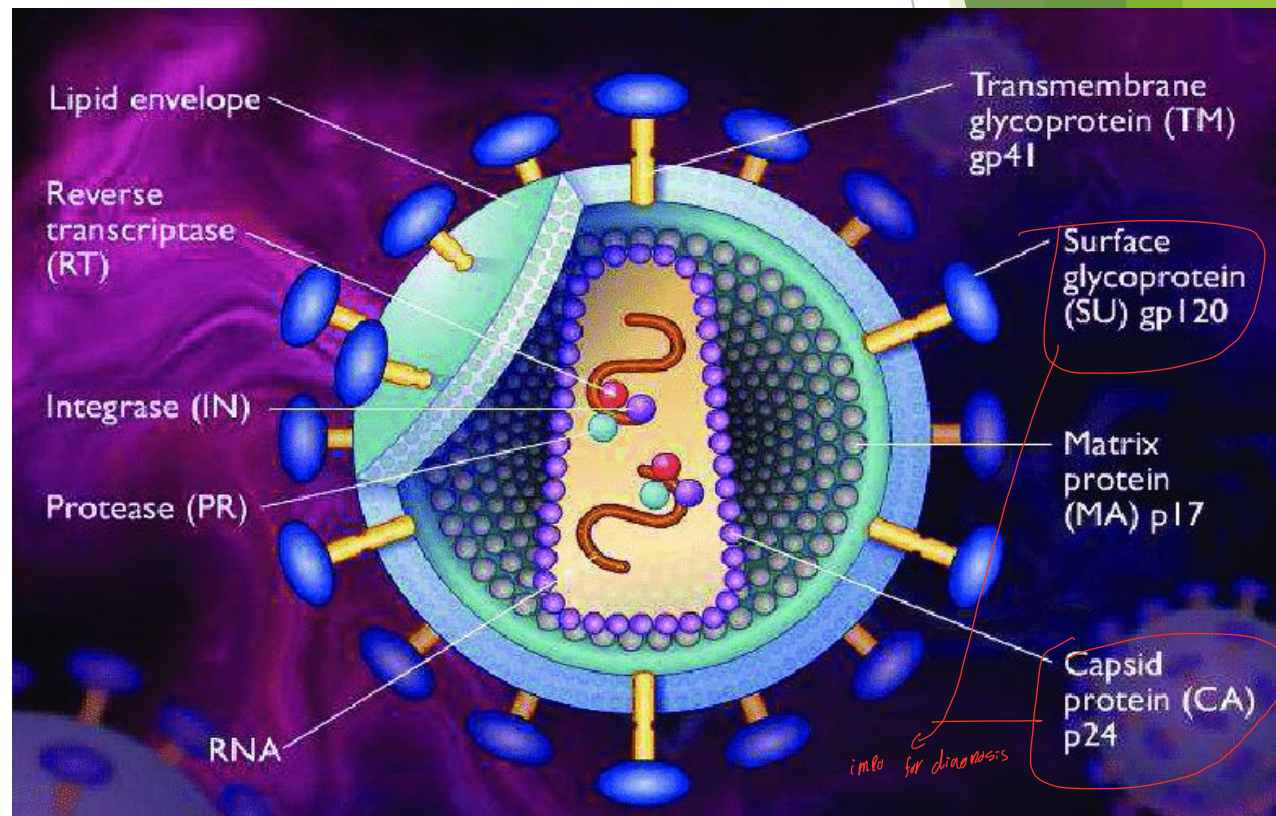


# Virology

- HIV is an enveloped ribonucleic acid (RNA) retrovirus from the lentivirus family
- Size:  $1/10000^{\text{th}}$  of millimeter ( 100 nanometer)
- It is a protein capsule containing two short stranded RNA and enzymes
- .

# Virology

- After mucosal exposure, HIV is transported via dendritic cells to the lymph nodes, where infection becomes established. This is followed by viremia and dissemination to lymphoid organs, which are the main sites of viral replication.



# Virology

- HIV infects cells bearing the **CD4 receptor which are:**
  - T-helper lymphocytes
  - Monocyte–macrophages
  - Dendritic cells and Microglial cells in the central nervous system (CNS).
- Entry into the cell commences with binding **of gp120 to the CD4 receptor** , which results in a conformational change in gp120 that permits binding to one of two chemokine co-receptors (**CXCR4 or CCR5**)

# Virology

- Individuals who are **homozygous for the CCR5 delta 32 mutation** do not express CCR5 on CD4 cells and are **immune to HIV infection**
- After penetrating the cell and uncoating, a deoxyribonucleic acid (DNA) copy is transcribed from the RNA genome by the reverse transcriptase enzyme, which is carried by the infecting virion.

# Virology

- Viral DNA is transported into the nucleus and integrated within the host cell genome by the integrase enzyme. Integrated virus is known as proviral DNA and persists for the life of the cell.
- Cells infected with proviral HIV DNA produce new virions only if they undergo cellular activation
- The mature virion infects other CD4 cells and the process is repeated.
- CD4 lymphocytes that are replicating HIV have a short survival time of about 1 day. *→ hence ↓ CD4+ TPR cell → CD4 dies new (replicated) virus move on and attack other CD4 and same scenario*

# Diagnosing HIV infection

- Globally, the trend is towards universal HIV testing, rather than testing only those patients at high risk or those with manifestations of HIV infection.  
*raising awareness so that everyone who testing → reducing social stigma*  
*if test + immediate start treatment*
- HIV is diagnosed by detecting host antibodies either with rapid point-of-care tests or in the laboratory, where enzyme-linked immunosorbent assay (ELISA) tests are usually done.
- Most tests detect antibodies to both HIV-1 and HIV-2.

# Diagnosing HIV infection

- Screening tests often include an assay for **p24 antigen** in addition to antibodies, in order to detect patients with primary infection before the antibody response occurs.

- **Two positive antibody tests from two different immunoassays are sufficient to confirm infection.**

\* but single antigen test is enough

# Diagnosing HIV infection

- Polymerase chain reaction (PCR) to detect HIV RNA are used to **diagnose infections in infants of HIV-positive mothers**, who carry maternal antibodies to HIV for up to 15 months irrespective of whether they are HIV-infected, and to diagnose primary infection before antibodies have developed.
- **PCR is more sensitive than p24 antigen detection for diagnosing primary infection**

↳ more sensitive than Ab

# Diagnosing HIV infection

- The purpose of HIV testing is not simply to identify infected individuals, but **also to educate people about prevention and transmission of the virus.**
- The extent of these investigations will depend on the resources available.

# CD4 count

*help us with assessing disease severity*

- The CD4 count is the **most clinically useful laboratory indicator of the degree of immune suppression**;
- it is used, together with **clinical staging**, in decisions to **start prophylaxis against opportunistic infections**
- It is of great value in the differential diagnosis of clinical problems. *eg like if patient present with ↓ CD4 we can probably say its HIV induced*

# CD4 count

- The CD4 count varies by up to 20% from day to day and is transiently reduced by intercurrent infections.
- The rate of decline in CD4 count is **highly variable**.
- **The normal CD4 count is over 500 cells/mm<sup>3</sup>.**
- People with CD4 counts between **200 and 500** cells/mm<sup>3</sup> have a low risk of developing major opportunistic infections.

# CD4 count

- Morbidity due to inflammatory **dermatoses, herpes zoster, oral candidiasis, tuberculosis, bacterial pneumonia** and HIV-related immune disorders (e.g. immune thrombocytopenia) becomes increasingly common as **CD4 counts decline**.
- Once the count is below 200 cells/mm<sup>3</sup>, there is severe immune suppression and a high risk of AIDS-defining conditions.
- It is important to note that patients can be asymptomatic despite very low CD4 counts and that major opportunistic diseases occasionally present with high CD4 counts.

↳ high, high rise of opportunistic disease

although usually absent in high CD4

# Viral load

- The level of viremia is measured by quantitative PCR of HIV RNA, known as the viral load.
- Determining the viral load is crucial for monitoring responses to ART .  
*also assess severity*  
*by*  
*antiretroviral - treatment*
- People with high viral loads (e.g.  $> 100\,000$  copies/mL) experience more rapid declines in CD4 count,

# Viral load and CD4 count

- while those with low viral loads ( $< 1000$  copies/mL) usually have slow or even no decline in CD4 counts.
- Viral loads are variable; **only changes** in viral load of more than  $0.5 \log_{10}$  copies/mL are considered clinically significant (i.e. 3 fold increase or decrease)

# Clinical manifestation of HIV

- Clinical staging of patients should be done at the initial medical examination, as it provides prognostic information and is a key criterion for initiating prophylaxis against opportunistic infections.
- **Two clinical staging systems** are used internationally  
WHO system and CDC system
- In both, patients are staged according to the most severe manifestation and do not improve their classification
- For example, a patient who is asymptomatic following a major opportunistic disease (AIDS) remains at stage 4 or category C of the WHO and CDC systems respectively, and never reverts to earlier stages.

# WHO clinical staging

## ► Stage 1

- Asymptomatic
- Persistent generalized lymphadenopathy

## Stage 2

- Unexplained moderate weight loss ( $< 10\%$  of body weight)
- Recurrent upper respiratory tract infections
- Herpes zoster, Angular cheilitis, Recurrent oral ulceration, *→ candida*
- Papular pruritic eruptions, Seborrhoeic dermatitis
- Fungal nail infections

# WHO clinical staging

## ► Stage 3 *the presence of one of the*

- Unexplained severe **weight loss** ( $> 10\%$  of body weight)
- Unexplained chronic **diarrhoea** for  $> 1$  month
- Unexplained persistent **fever** ( $> 37.5^{\circ}\text{C}$  for  $> 1$  month)
- Persistent **oral candidiasis**, Oral hairy leucoplakia
- Pulmonary <sup>*(reactivated)*</sup> **tuberculosis**, Severe bacterial infections
- Acute necrotizing **ulcerative stomatitis**, gingivitis or periodontitis
- Unexplained **anemia** ( $< 80$  g/L), neutropenia ( $< 0.5 \times 10^9/\text{L}$ ) and/or chronic thrombocytopenia ( $< 50 \times 10^9/\text{L}$ )

# WHO clinical staging

## Stage 4

- Candidiasis of esophagus, trachea, bronchi or lungs
- Cervical carcinoma – invasive Cryptococcosis – extrapulmonary,
- Cryptosporidiosis, chronic ( $> 1$  month)
- Cytomegalovirus disease (outside liver, spleen and nodes)
- Herpes simplex chronic ( $> 1$  month) ulcers or visceral
- HIV encephalopathy, HIV wasting syndrome

# WHO clinical staging

## ► Stage 4

- Cystoisosporiasis , Kaposi's sarcoma, Lymphoma, Mycobacterial infection, non-tuberculous, extrapulmonary or disseminated
- Mycosis – disseminated endemic
- Pneumocystis pneumonia
- Pneumonia, recurrent bacterial
- Progressive multifocal leukoencephalopathy
- Toxoplasmosis – cerebral
- Tuberculosis – extrapulmonary (CDC includes pulmonary)
- Sepsis

# Clinical manifestation of HIV

Patients do not always progress steadily through all stages and may present with AIDS, having been asymptomatic.

**Primary HIV infection:** it is symptomatic in more than 50% of cases but the diagnosis is often missed.

- The incubation period is usually 2–4 weeks after exposure. The duration of symptoms is variable but is seldom longer than 2 weeks.
- The clinical manifestations **resemble those of infectious mononucleosis**, but the presence of **maculopapular rash** or **mucosal ulceration** strongly suggests primary HIV infection rather than the other viral causes of infectious mononucleosis, in which rashes generally occur only if aminopenicillins are given.

# Clinical features of primary infection

- Fever
- Maculopapular rash
- Pharyngitis
- Lymphadenopathy
- Myalgia/arthralgia
- Diarrhea
- Headache
- Oral and genital ulceration
- Meningo-encephalitis
- Bell's palsy

# Clinical manifestation of HIV

- Atypical lymphocytosis occurs less frequently than in Epstein–Barr (EBV) infection.
- Transient lymphopenia, including CD4 lymphocytes, is found in most cases , which may result in opportunistic infections, notably oropharyngeal candidiasis.
- Major opportunistic infections like Pneumocystis pneumonia may rarely occur.
- Thrombocytopenia and moderate elevation of liver enzymes are commonly present.

# Clinical manifestation of HIV

- The differential diagnosis of primary HIV includes :
  - Acute EBV,
  - Primary cytomegalovirus (CMV) infection,
  - Rubella,
  - Primary toxoplasmosis
  - Secondary syphilis.
- Early diagnosis is made by detecting HIV RNA by PCR or p24 antigenemia.  
*↳ 2 weeks*
- The appearance of specific anti-HIV antibodies in serum occurs 2–12 weeks after the development of symptoms.

*↳ do it if suspecting with -ve Ab do recurrent test 12 weeks later*

# Reservoir of infection

- ▶ Cases and carriers
- ▶ Once a person is infected , virus remain lifelong
- ▶ It can be transmitted even if the patient is symptomless

# Modes of transmission

HIV is transmitted by :

- **Sexual contact**, *→ from male to female ≥ female to male*
- **Exposure to blood** (e.g. injection, drug use, occupational exposure in health-care workers) and blood products,
- **Infants of HIV-infected mothers**, who may be infected in utero (this is uncommon), perinatally or via breastfeeding *→ low risk*  
*→ economic problem*

*so if there's high risk of malnutrition by stopping breastfeeding you can do risk-benefit assessment*

# Modes of transmission

- The risk of contracting HIV after exposure to infected body fluid is dependent on :
  - The integrity of the exposed site,
  - the type and volume of fluid,
  - the level of viremia in the source person.  
*(viral load)*

# Modes of transmission

- A high proportion of patients with haemophilia in high-income countries were infected through contaminated blood products before HIV antibody screening was adopted in 1985.
- Routine screening of blood and blood products for HIV infection has virtually eliminated this mode of transmission
- However, because of the lack of adequate screening facilities in resource-poor countries, it has been estimated that 5%–10% of blood transfusions globally are with HIV-infected blood.

# HIV exposure Approximate risk

JK

## HIV exposure

## Approximate risk\*

### Sexual

also risk is low but since they do it multiple times  
hence risk cumulative

Vaginal intercourse: female to male

0.05%

Vaginal intercourse: male to female

0.10%

Anal intercourse: insertive

0.10%

Anal intercourse: receptive

1.40%

Oral intercourse: insertive

< 0.01%

Oral intercourse: receptive

< 0.01%

### Blood exposure

Blood transfusion

90%

Intravenous drug-users sharing needles

0.67%

Percutaneous needlestick injury

0.30%

Mucous membrane splash

0.09%

### Mother to child

Vaginal delivery

15%

Breastfeeding (per month)

0.50%

\* These are approximate estimates and assume that the source person is not being treated with ART.

# Factors increasing the risk of transmission of HIV

## 1- Common to all transmission categories:

- High viral load

## 2- Sexual transmission

- STIs, especially genital ulcers  
*→ always screen for Hep-B, Hep-C, HSV*
- Cervical ectopy
- Rectal or vaginal lacerations
- Menstruation
- Uncircumcised male partner
- Receptive anal intercourse
- Sexual activity with recreational drug use ('chemsex')  
*→ unaware what they are doing  
(safe, multiple, type?)*



# Factors increasing the risk of transmission of HIV

## 3- Injection drug use transmission

- Sharing equipment
- Linked commercial sex
- Intravenous use
- Concomitant cocaine use
- Incarceration ( imprisonment)

# Factors increasing the risk of transmission of HIV

## 4- Occupational transmission

- Deep injury
- Visible blood on device
- Needle was in a blood vessel

## 5- Vertical transmission:

Prolonged rupture of membranes

- Older gestational age

# Minor HIV-associated disorders

- A wide range of disorders indicating some impairment of cellular immunity occur in most patients before they develop AIDS (CDC category B or WHO stages 2 and 3).
- Careful examination of the mouth is important
- when patients are being followed up, as oral candidiasis and oral hairy leucoplakia are common conditions that require initiation of prophylaxis against opportunistic infections, irrespective of the CD4 count.

*White patch on mucous membrane*

# Acquired immuno-deficiency syndrome

- ▶ AIDS is defined by the development of specified opportunistic infections, cancers and severe manifestations of HIV itself (CDC category C or WHO stage 4).

# Treatment

- The treatment for HIV is called antiretroviral therapy (ART)
- ART involves taking a combination of HIV medicines (called an HIV treatment regimen) every day *usually 3 drugs of 2 types*
- ART is recommended for everyone who has HIV.
- People with HIV should start taking HIV medicines as soon as possible
- ART cannot cure HIV, but HIV medicines help people with HIV live longer, healthier lives. ART also **reduces the risk** of HIV transmission *Reduce opportunistic infection*

*Prevents  
(2010s)*

# Treatment

- A main goal of HIV treatment is to reduce a person's viral load to an undetectable level.
- An undetectable viral load An undetectable viral load means that the level of HIV in the blood is too low to be detected by a viral load test.
- People with HIV who maintain an undetectable viral load have effectively no risk of transmitting HIV to their HIV-negative partners through sex.

leads to ↓ CD4  
↓ opportunistic infections  
↓ risk of transmission



# Drug classes *memo: sust the red/bold*

1. **Non-nucleoside reverse transcriptase inhibitors (NNRTIs):** as **efavirenz**, rilpivirine and doravirine.
2. **Nucleotide reverse transcriptase inhibitors (NRTIs) :** as abacavir , **tenofovir disoproxil fumarate** , **emtricitabine**, **lamivudine** and zidovudine . **Combination drugs also are available**, such as emtricitabine/tenofovir disoproxil fumarate and emtricitabine/tenofovir alafenamide fumarate.
3. **Protease inhibitors (PIs):** as atazanavir, darunavir and lopinavir/ritonavir .
4. **Integrase inhibitors :**as bictegravir sodium/emtricitabine/tenofovir alafenamide fumarate, raltegravir, dolutegravir and cabotegravir.
5. **Entry or fusion inhibitors:** as **enfuvirtide** and maraviroc

# HIV Prevention (tips)

- 1. Protection during sex:** abstinence (not having sex), limit number of partners or use condom during every sex
- 2. Protection during drug injection:** drug dealers should use new, clean syringes and injection equipment every time and never share a needle.
- 3. Prevent perinatal transmission:** the earlier HIV is diagnosed and treated, the more effectively HIV medicine will reduce viral load prevent transmission trans-placentally. *also better to avoid breast-feeding*
- 4. PEP :** *Post-exposure - PRO PHARMS* should be used with 72 hours of exposure, **Tenofovir together with emtricitabine** is the most widely used dual NRTI combination together with either a PI or an integrase inhibitor
- 5. PrEP :** *Pre-exposure - PRO PHARMS* daily tenofovir plus emtricitabine has been shown to reduce the risk of HIV acquisition in people who are at ongoing high risk (e.g. from sex or injecting drug use) and is well tolerated. Regular HIV testing should be done in people on PrEP.

*\* Confidentiality in management but suspected & confirmed case → report to CDC*